

Calculus I Lesson 26

1) $f(x) = 3x^2 + 3x - 5$, $x \in [-2, 1]$

Let $a = -2$ and $b = 1$

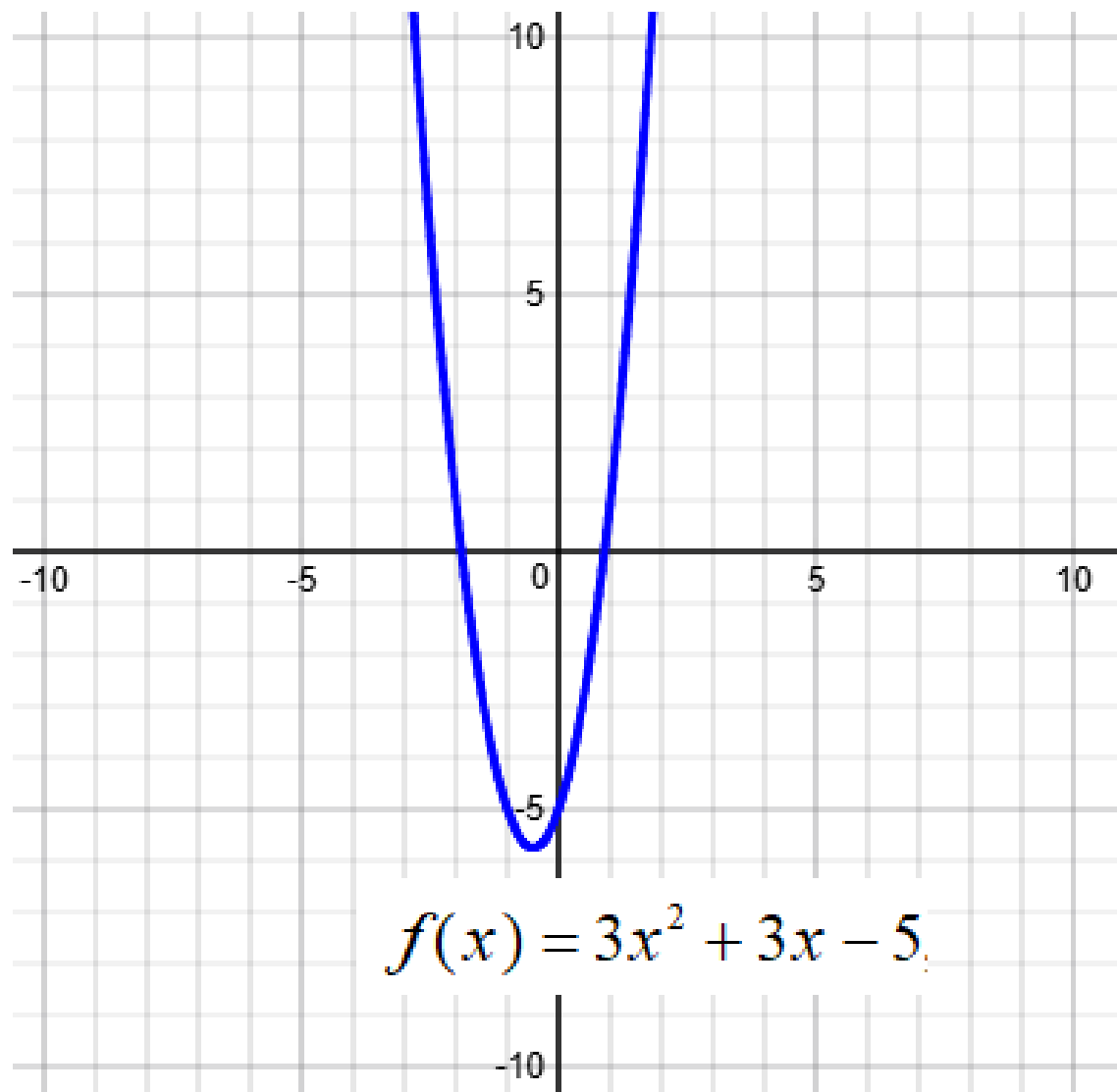
a) $f(a) = \underline{\quad ? \quad}$

b) $f(b) = \underline{\quad ? \quad}$

- c) Rolle's Theorem says that if a function $f(x)$ is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , and $f(a) = f(b)$, then there must be at least one number $c \in (a, b)$ such that $f'(c) = 0$.

Find c such that $f'(c) = 0$.

Hint: Find $f'(x)$, set $f'(x) = 0$, and solve for x .



2) $f(x) = -x^2 + 8x - 5, \quad x \in [0, 8]$

Let $a = 0$ and $b = 8$

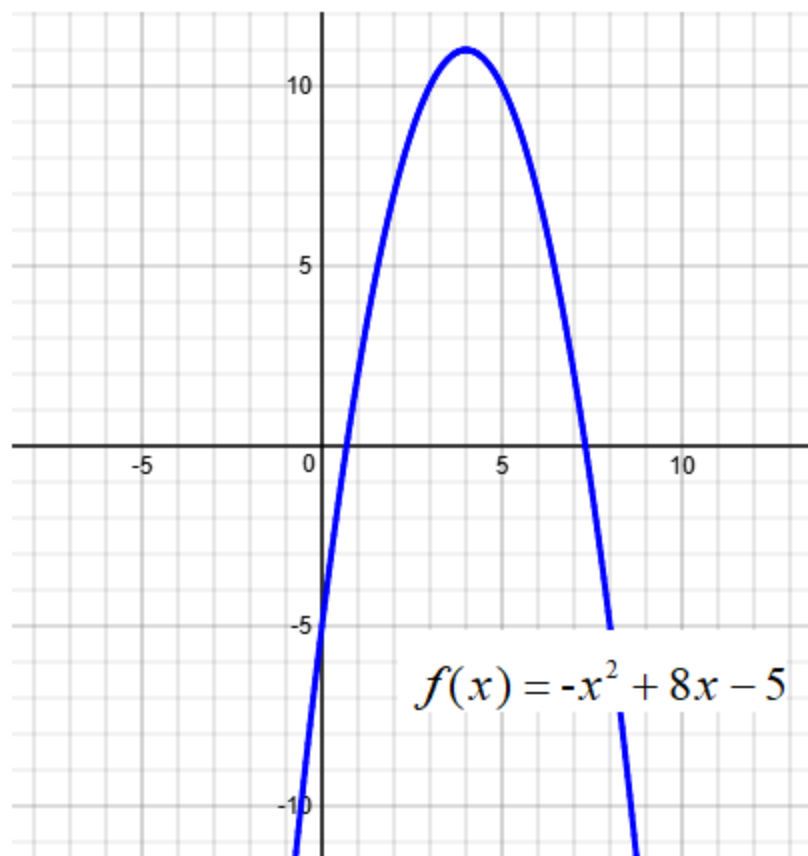
a) $f(a) = \underline{\quad ? \quad}$

b) $f(b) = \underline{\quad ? \quad}$

c) Rolle's Theorem says that if a function $f(x)$ is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , and $f(a) = f(b)$, then there must be at least one number $c \in (a, b)$ such that $f'(c) = 0$.

Find c such that $f'(c) = 0$.

Hint: Find $f'(x)$, set $f'(x) = 0$, and solve for x .



3) $f(x) = x^3 - 2x^2 - 5x + 6, \quad x \in [1, 3]$

Let $a = 1$ and $b = 3$

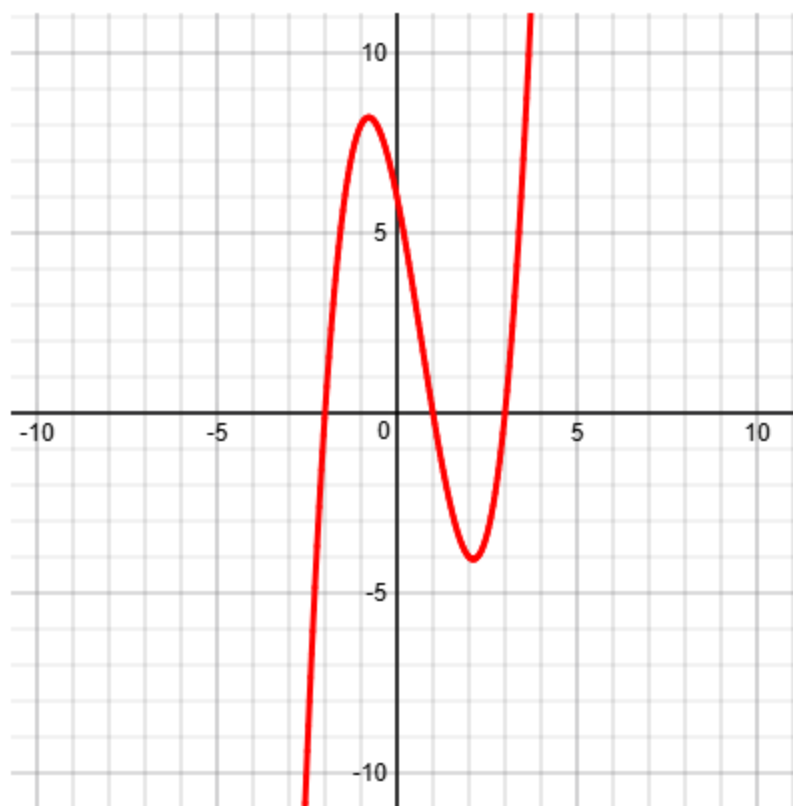
a) $f(a) = \underline{\quad ? \quad}$

b) $f(b) = \underline{\quad ? \quad}$

c) Rolle's Theorem says that if a function $f(x)$ is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , and $f(a) = f(b)$, then there must be at least one number $c \in (a, b)$ such that $f'(c) = 0$.

Find c such that $f'(c) = 0$.

Hint: Find $f'(x)$, set $f'(x) = 0$, and solve for x .



$$f(x) = x^3 - 2x^2 - 5x + 6$$

4) $f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$, $x \in [-3, -2]$

Let $a = -3$ and $b = -2$

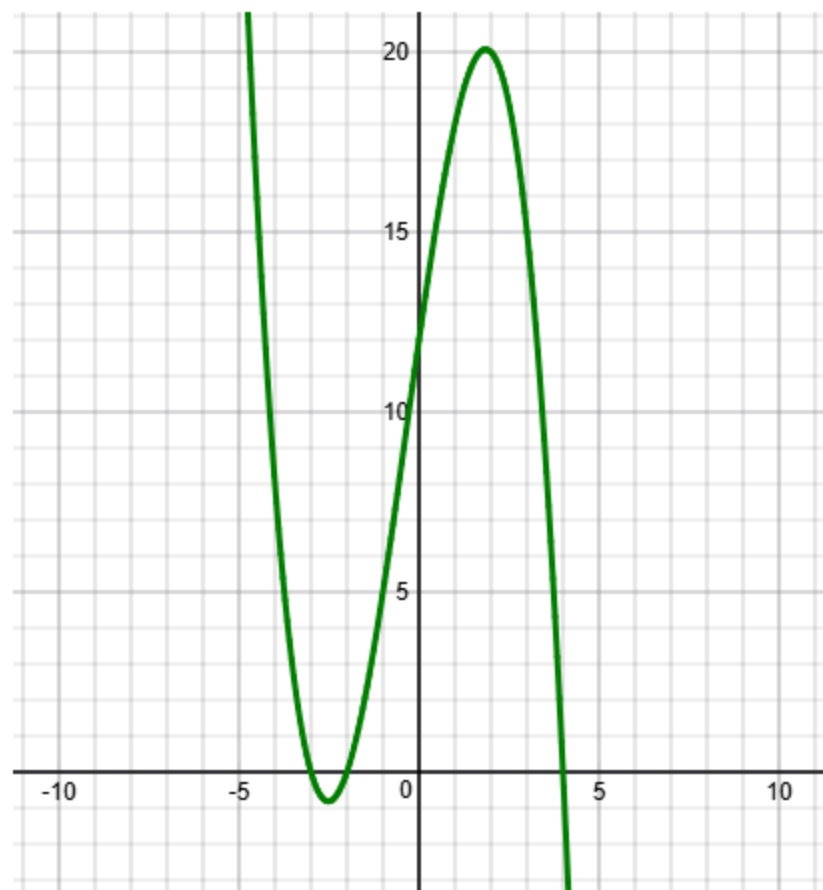
a) $f(a) = \underline{\quad ? \quad}$

b) $f(b) = \underline{\quad ? \quad}$

c) Rolle's Theorem says that if a function $f(x)$ is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , and $f(a) = f(b)$, then there must be at least one number $c \in (a, b)$ such that $f'(c) = 0$.

Find c such that $f'(c) = 0$.

Hint: Find $f'(x)$, set $f'(x) = 0$, and solve for x .



$$f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$$

5) $f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$, $x \in [-2, 1]$

Let $a = -2$ and $b = 1$

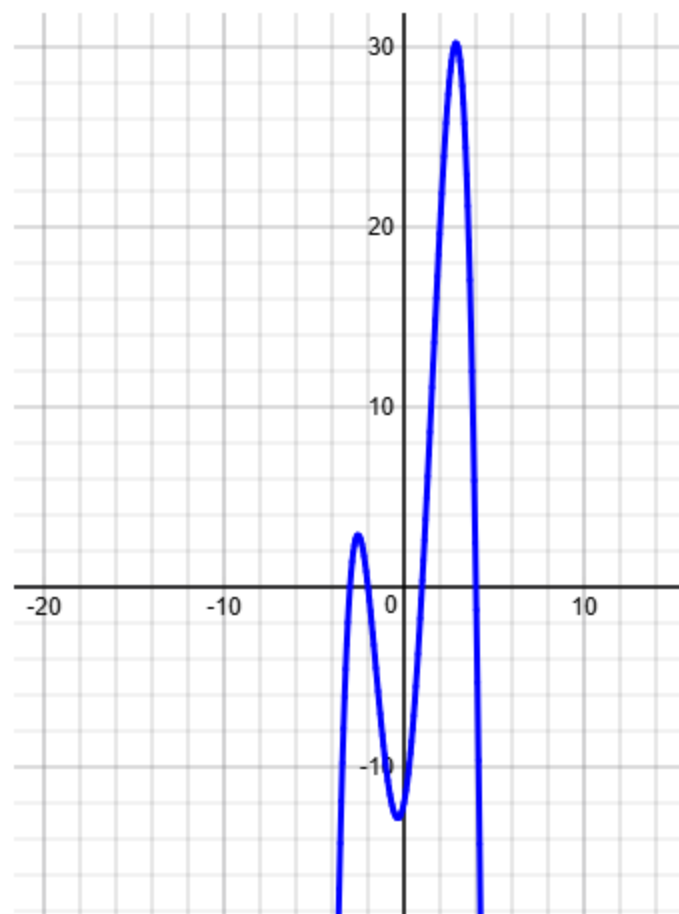
a) $f(a) = \underline{\hspace{1cm}}?$

b) $f(b) = \underline{\hspace{1cm}}?$

c) Rolle's Theorem says that if a function $f(x)$ is continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b) , and $f(a) = f(b)$, then there must be at least one number $c \in (a, b)$ such that $f'(c) = 0$.

Find c such that $f'(c) = 0$.

Hint: Find $f'(x)$, set $f'(x) = 0$, and solve for x .



$$f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$$

